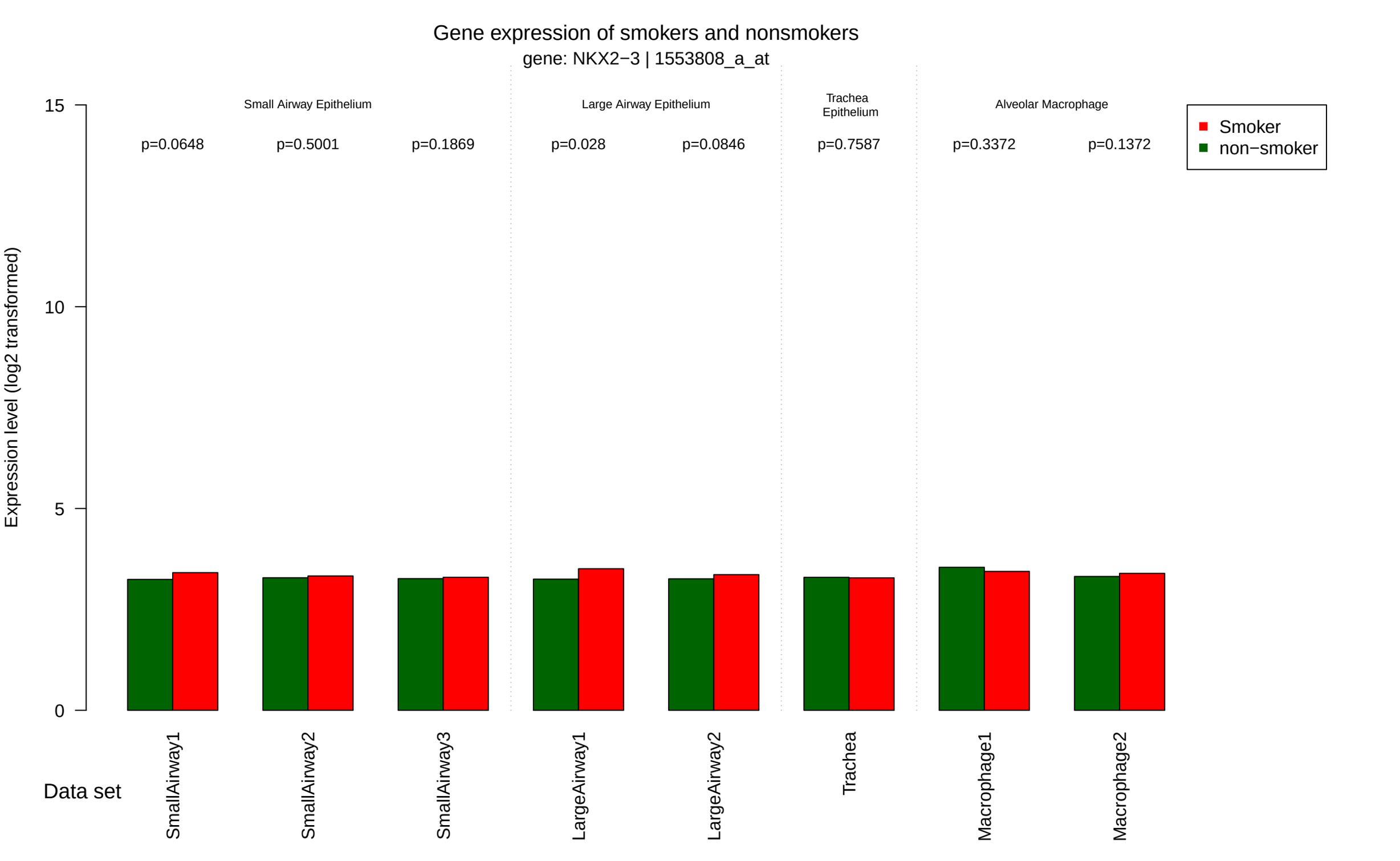
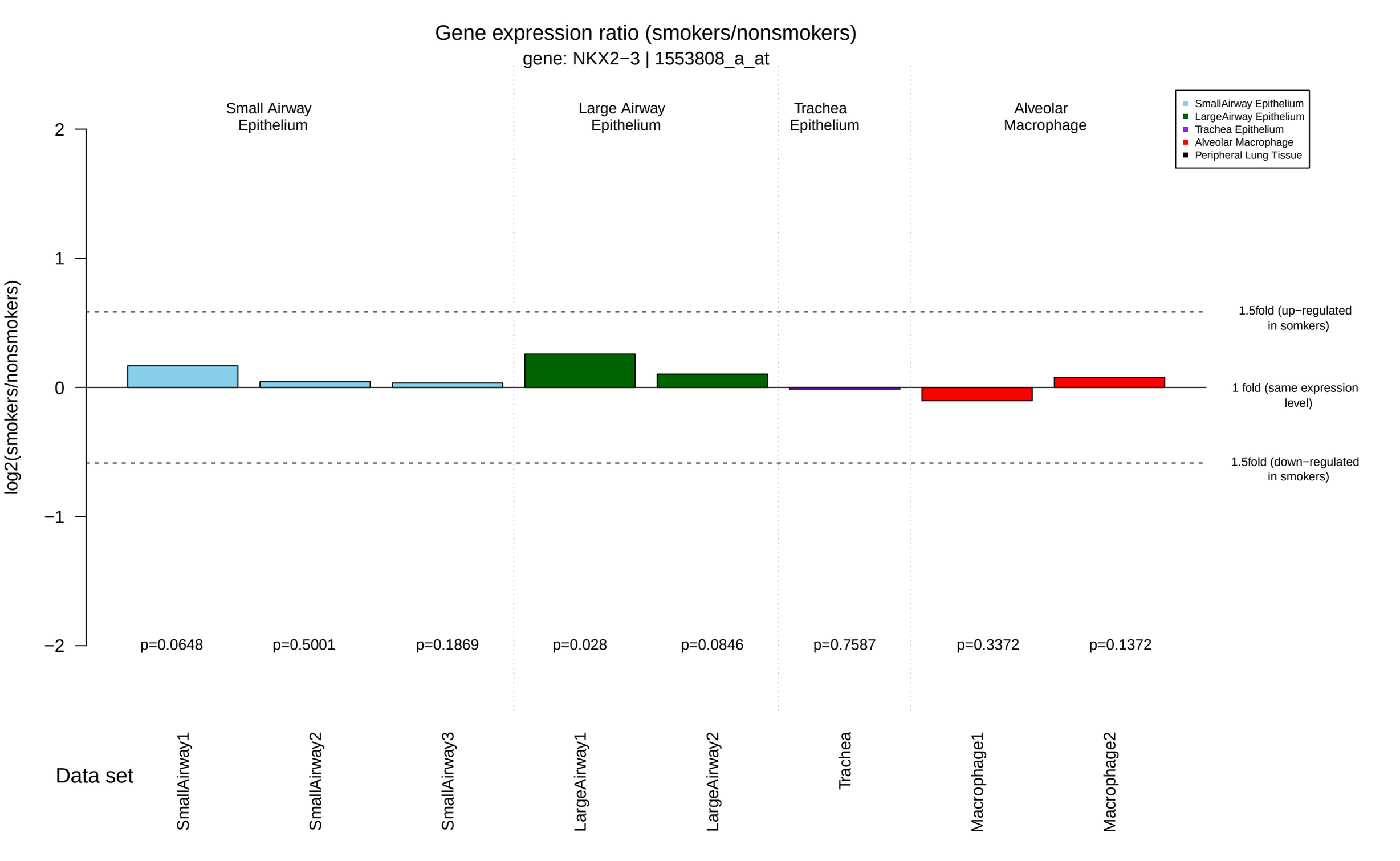
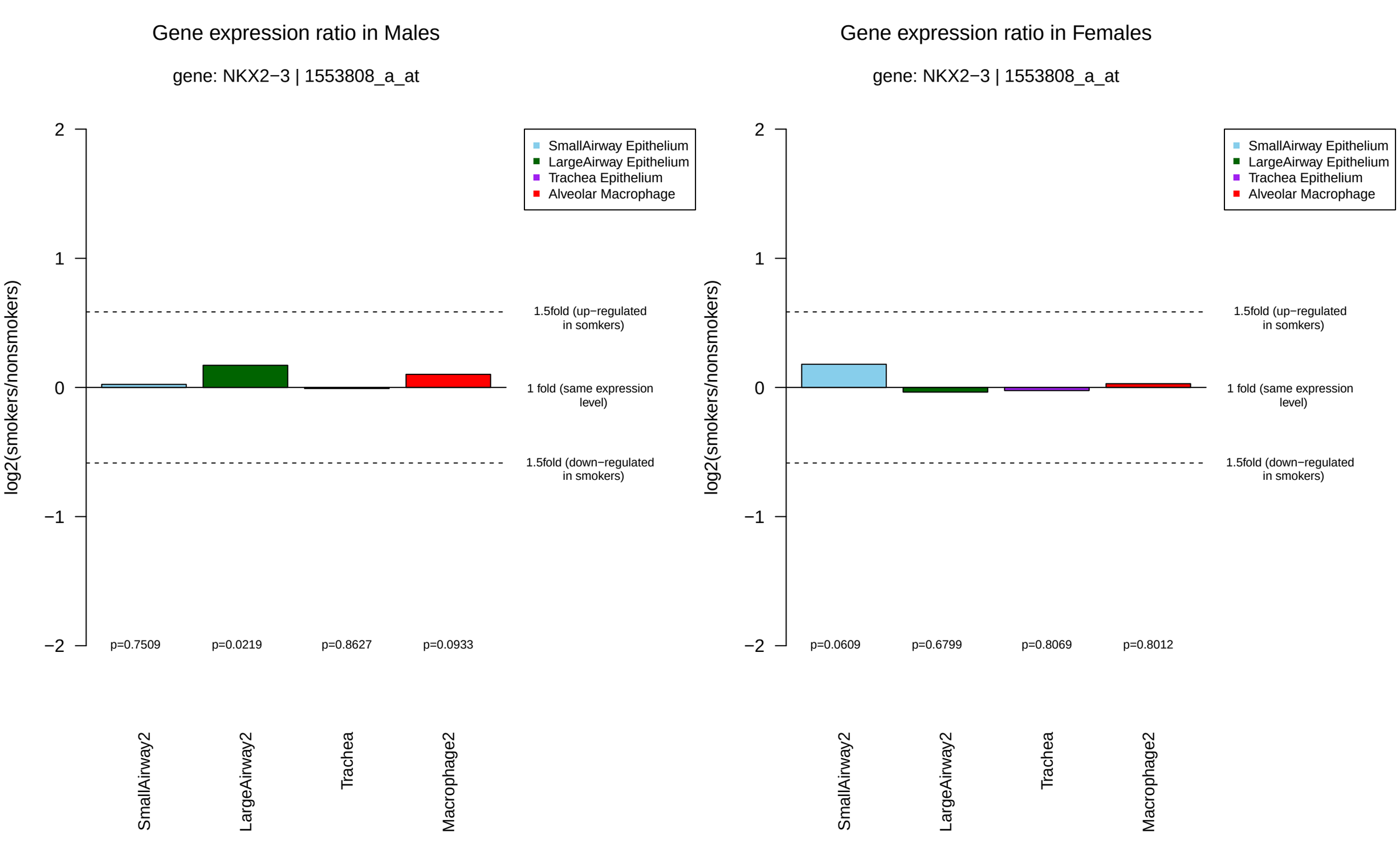
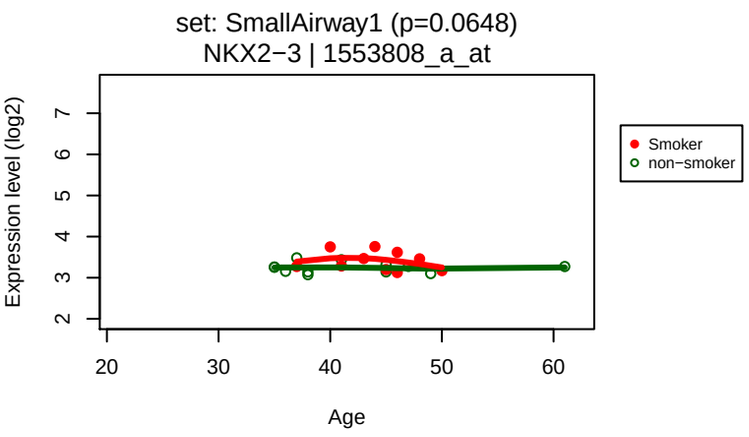
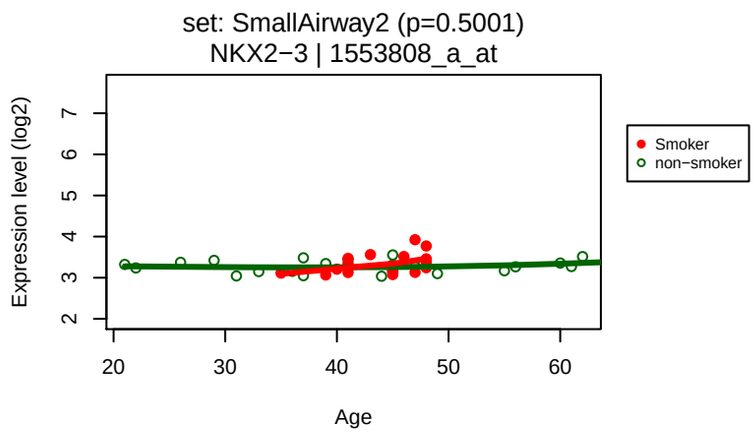
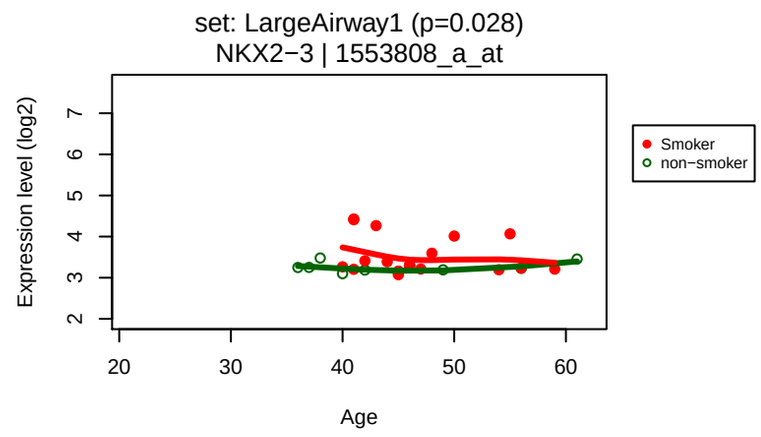

Gene expression of smokers and nonsmokers
gene: NKX2−3 | 1553808_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0648
p=0.5001
p=0.1869
p=0.028
p=0.0846
p=0.7587
p=0.3372
p=0.1372
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

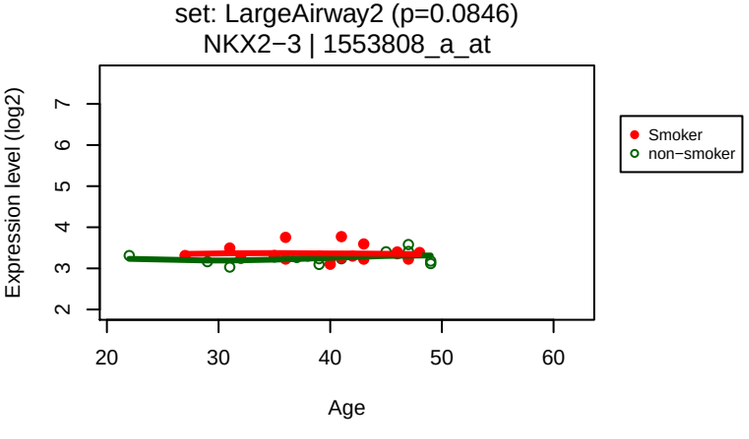

Gene expression ratio (smokers/nonsmokers)
gene: NKX2−3 | 1553808_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0648
p=0.5001
p=0.1869
p=0.028
p=0.0846
p=0.7587
p=0.3372
p=0.1372
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

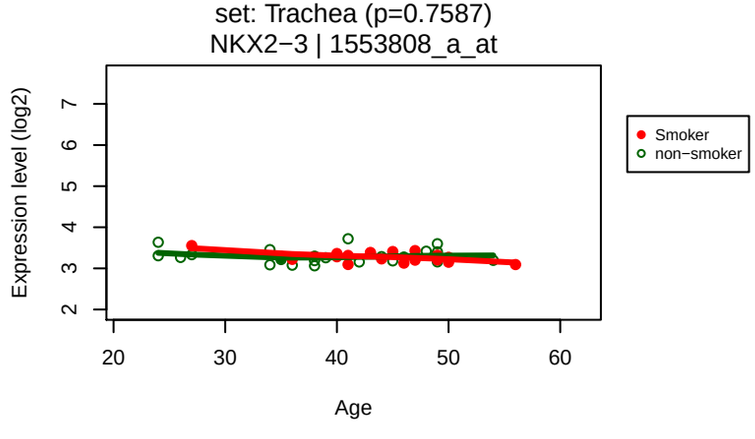

Gene expression ratio in Males
gene: NKX2−3 | 1553808_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7509
p=0.0219
p=0.8627
p=0.0933
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NKX2−3 | 1553808_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0609
p=0.6799
p=0.8069
p=0.8012
SmallAirway2
LargeAirway2
Trachea
Macrophage2

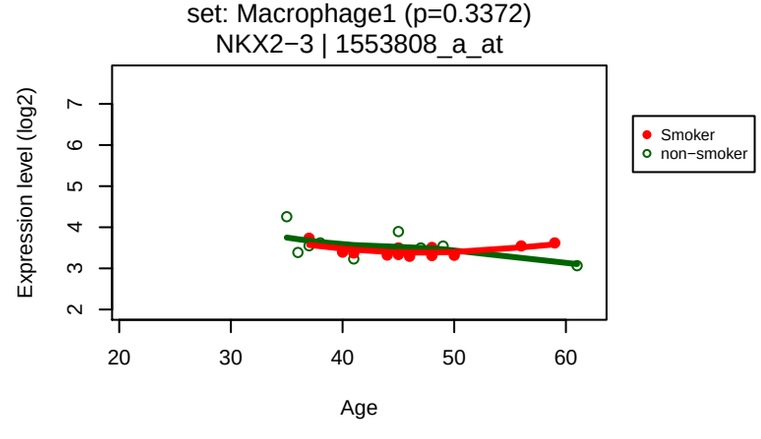

set: SmallAirway1 (p=0.0648)
NKX2−3 | 1553808_a_at
Expression level (log2)
Age
Smoker
non−smoker

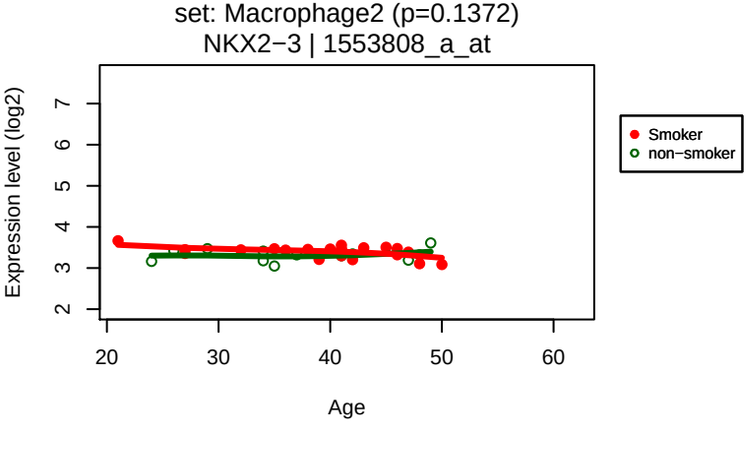

set: SmallAirway2 (p=0.5001)
NKX2−3 | 1553808_a_at
Expression level (log2)
Age
Smoker
non−smoker

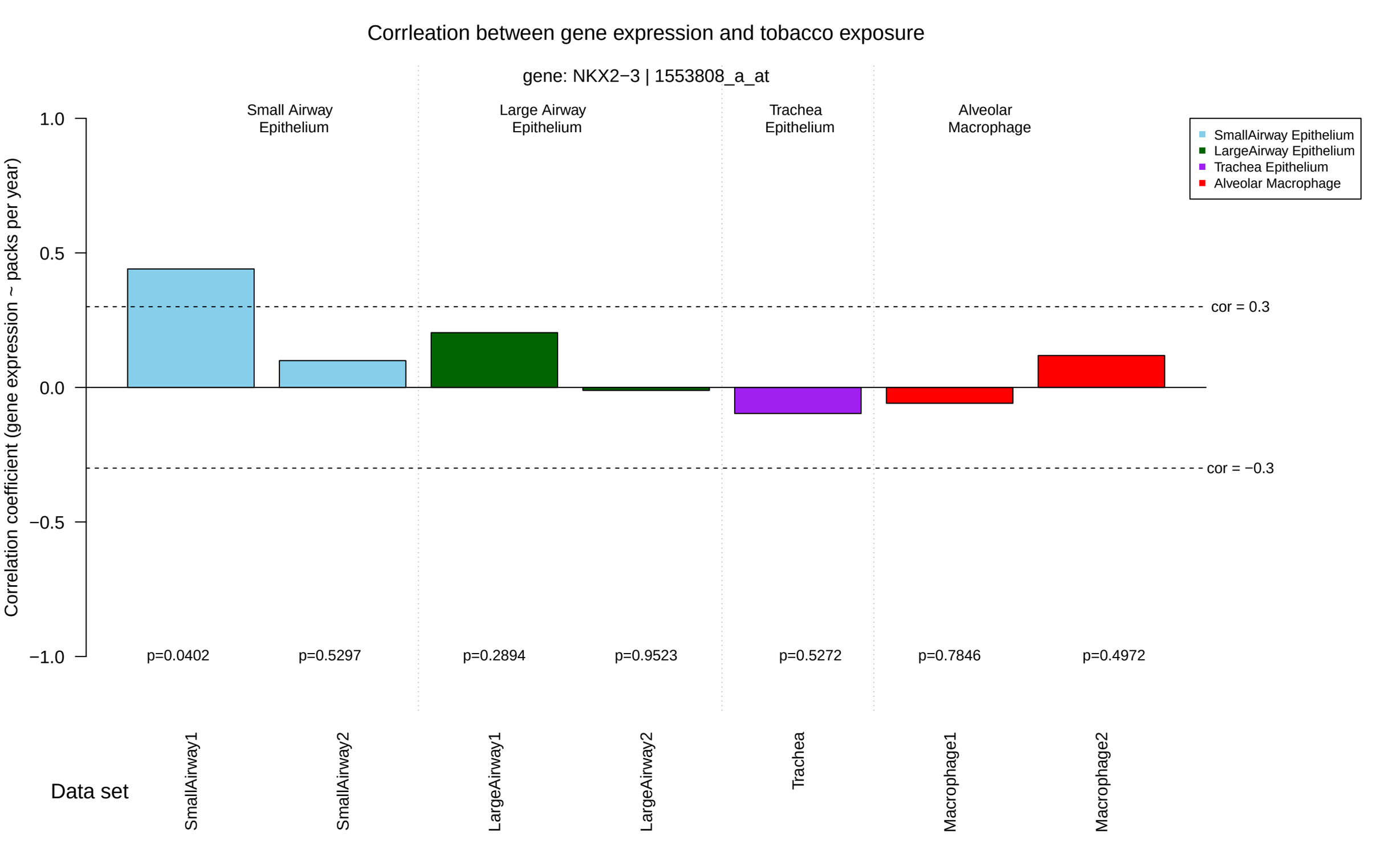

set: LargeAirway1 (p=0.028)
NKX2−3 | 1553808_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0846)
NKX2−3 | 1553808_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7587)
NKX2−3 | 1553808_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3372)
NKX2−3 | 1553808_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1372)
NKX2−3 | 1553808_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NKX2−3 | 1553808_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0402
p=0.5297
p=0.2894
p=0.9523
p=0.5272
p=0.7846
p=0.4972
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2